

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-YD-2301-04 Sheet 1 of 1
OPL Title	Controlled Temperature Ramp-Up Procedure	Discipline	Process / Operations	
Equipment	YD-2301 - POLYMER FLUID BED DRYER	Area / Unit	2300 - POWDER DRYING SYSTEM	
Related Interlock	SEQ-5500 (SIL 2)	P&ID; Ref	TJC-LLD-PID-2301	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **controlled temperature ramp-up procedure** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. Heating the drum too quickly warps the shell and cracks the steam tubes, causing tube leaks and long downtime. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Ramp TIC-2301 at a maximum of 10 degC per 10 minutes	Permit valid & briefed
2	Hold at 60 degC to let the drum warm through evenly	Parameter within limit
3	Watch drum thermal growth at the trunnion supports	Interlock status confirmed
4	Reach the 95-110 degC operating band slowly	Change logged on DCS
5	If uneven expansion / rubbing is heard, pause the ramp	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
Scheduled proof test of dryer outlet high-high temperature trip	Set-point within tolerance, minor zero drift corrected	Recalibrated TSHH-2301 to 125 degC and function tested to logic

6. KEY LEARNING POINTS

- Heating the drum too quickly warps the shell and cracks the steam tubes, causing tube leaks and long downtime.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.